Before You Press the Button
Even when the PC is OFF but **plugged into power**, some parts of the motherboard are still ON.

✅ **Important Term:**

- **+5VSB (5 Volt Standby Power)**
  This is always ON when the computer is plugged in. It powers a special chip called the **EC (Embedded Controller)** or **Super I/O (Input/Output Controller)**.

⭕ **1. You Press the Power Button**
What Happens?
- The **Power Button** sends a signal to the **EC (Embedded Controller)** or **Super I/O chip**.

Signal:
- **PWRBTN# (Power Button)** – this tells the system to turn ON.

⚡ **2. EC or Super I/O Says: "Let's Power Up!"**
What Happens?
- EC or Super I/O sends a signal to the **PCH (Platform Controller Hub)** to begin turning ON the full system.

Signals:
- **PCH_ON# (Platform Controller On)**
- Then, **PCH** sends **PS_ON# (Power Supply On)** to the **PSU (Power Supply Unit)**.

🔌 **3. Power Supply Turns ON**
What Happens?
- The **PSU** now turns on the **main power rails**:
  - +12V
  - +5V
  - +3.3V

Then it sends back a signal:
- **PWR_OK (Power OK)**
  This tells the motherboard that all voltages are **stable**.

🔧 **4. Motherboard Chips Start Working**
Voltage Regulation:
- Special circuits called **VRMs (Voltage Regulator Modules)** turn +12V into smaller voltages like:
  - **VCORE (CPU Core Voltage)**
  - **VDDQ (Memory Voltage)**

These are sent to:
- **CPU (Central Processing Unit)**
- **RAM (Random Access Memory)**

⏱️ **5. Clock & Reset Signals Start**
Clock Signals:

- The **Clock Generator** chip starts sending timing signals (like a heartbeat).

Reset Signal:

- **RESET#** is removed — allowing chips to start working.

🧠 6. CPU Starts Thinking

What Happens?

- The **CPU** starts working and looks for instructions in the **BIOS chip** (a small memory chip).

Dell> Boot Device> HDD 1 > Boot MGR

BIOS Full Form:

- **Basic Input/Output System**

The CPU runs the BIOS to check if everything is okay.

🧪 7. BIOS Runs POST

POST Full Form:

- **Power-On Self Test**

BIOS checks:

- Is RAM working?
- Is the display connected?
- Are hard drives found?

If something is wrong, it gives **beep codes** or shows errors on screen.

💻 8. Booting the Operating System

If everything is okay:

- BIOS gives control to a program on your SSD/HDD (like Windows bootloader).
- The operating system loads, and your PC is ready!

📡 Signal Summary Table (with Full Forms)

Signal/Chip	Full Form	Job
PSU	Power Supply Unit	Gives power to motherboard
EC	Embedded Controller	Manages power button, battery (in laptops)
Super I/O	Super Input/Output Controller	Manages fan, keyboard, ports (in desktops)
PCH	Platform Controller Hub	Controls RAM, storage, USB, etc.
VRM	Voltage Regulator Module	Converts 12V to smaller voltages
BIOS	Basic Input/Output System	Starts the system and checks hardware
CPU	Central Processing Unit	The brain of the computer
POST	Power-On Self Test	Hardware check before booting
RAM	Random Access Memory	Temporary memory for CPU
PWRBTN#	Power Button (active low signal)	Starts the system
PS_ON#	Power Supply On	Tells PSU to turn on
PWR_OK	Power OK	PSU says power is stable
RESET#	Reset Signal	Starts the chips

CLK	Clock Signal	Keeps all chips in sync

✅ **Example Summary: "From Button to Boot"**

1. Plug in → 5VSB powers EC.
2. Press button → EC sends signal.
3. PSU turns on full power.
4. VRMs power CPU and RAM.
5. BIOS runs POST.
6. CPU loads Windows.